Antra

Software Engineer | Data Scientist
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Summary

Computer Science and Engineering (Data Science) undergraduate with production experience in full stack web development and applied machine learning. Delivered role-based access control for a live HRMS platform, real-time WebSocket trading systems, and biometric ML applications. Merged 36 pull requests across 20 open-source repositories. Proficient in Python, TypeScript, React, Next.js, FastAPI, Node.js, SQL, and Scikit-learn.

Technical Skills

Languages: Python, JavaScript (ES6+), TypeScript, Java, C, C++, SQL
Frontend: React 18, Next.js, Vite, HTML5, CSS3, Tailwind CSS, shadcn/ui, Radix UI, React Hook Form, Zod
Backend: Node.js, Express.js, FastAPI, REST APIs, WebSockets (Socket.io), JWT Authentication, RBAC
Databases: PostgreSQL, MongoDB, MySQL
Cloud & Deployment: Render, Vercel, GitHub Actions, CI/CD Pipelines, Serverless Deployment
Data & ML: NumPy, Pandas, Scikit-learn, Matplotlib, OpenCV, MediaPipe, dlib, Resemblyzer, Librosa
AI and LLMs: Gemini 2.5 Flash, Groq, Prompt Engineering, Structured Output Parsing, OpenAPI Generation
Tools: Git, GitHub, Postman, npm, Figma, Agile Development, Unit Testing

Experience

Full Stack Developer Intern May 2026 – Jul 2026
SmartSkale *Remote*

- Built role-based access control (RBAC) across 2 production HRMS portals using React 18, TypeScript, and FastAPI, delivering page-level permission mapping, multi-role assignment, and edit versus view access resolution.
- Delivered 4 user management workflows covering role activation, role deactivation, audit logging, and password reset, shipping 5 merged pull requests into the production codebase.

Open Source Contributor May 2026 – Aug 2026
GirlScript Summer of Code (GSSoC) *Remote*

- Merged 36 pull requests across 20 repositories, covering bug fixes, feature development, and test coverage in React and Next.js frontends and Python and FastAPI backends.
- Restored a 454-line backend AuthService in story-spark-ai (1,000+ forks) after a merge conflict silently removed it, and resolved two additional latent authentication defects.
- Developed NyayaVanni, a Manifest V3 Chrome extension for AI-powered legal document analysis, resolving cross-origin session authentication and Gemini SDK schema compatibility issues.

Co-Creative Head 2026 – Present
ISTE VIT Student Chapter *Vellore, India*

- Lead an 8-member team driving technical events and content production for one of the largest student chapters at VIT.

Projects

PosePulse | *Next.js, FastAPI, WebSockets, MediaPipe, OpenCV, Groq LLM* GitHub | Live

- Built a real-time AI fitness coach that streams webcam frames over WebSockets for MediaPipe pose detection across 5 exercises, with repetition and set counting plus form correction logic.
- Added LLM-generated spoken coaching cues with cooldown gating, deploying the FastAPI backend on Render and the Next.js frontend on Vercel.

Snapclass | *Python, Streamlit, dlib, Resemblyzer, Scikit-learn* GitHub | Live

- Developed a dual-modal biometric attendance system pairing facial recognition (dlib with an SVM classifier) and voice verification using Resemblyzer speaker embeddings.
- Shipped role-based teacher and student dashboards with authenticated sessions, data sync, and QR-based course enrollment.

TradeDash | *React, Node.js, Express, Socket.io, MongoDB, TradingView* GitHub | Live

- Built a real-time US equity trading terminal with Finnhub WebSocket price streaming and TradingView candlestick and volume charts.
- Implemented JWT-protected REST endpoints with TTL caching, supporting MARKET, LIMIT, and STOP_LOSS order execution on both buy and sell sides.

CreditWise Loan System | *Python, Pandas, NumPy, Scikit-learn, Matplotlib* GitHub

- Trained and compared KNN, Logistic Regression, and Naive Bayes classifiers on 4,269 loan records, reaching 92% accuracy with Logistic Regression scored on precision, recall, and F1.

Education

Vellore Institute of Technology 2024 – 2028
B.Tech in Computer Science and Engineering with Data Science *CGPA: 8.89 / 10*

Patents

Microcontroller Synchronised Access Protocol via Fixed-Lattice Input with Coupled Electromechanical and Lumino-Acoustic Feedback, Co-Inventor, Indian Patent Office (IN202641047706), Published April 2026.